Unpatented Innovation: Evidence from Firms' Open Source Contribution

Shen Zhang

@ Stillman Research Day

Open-Source Software is Non-trivial

"Software is eating the world."

- Mark Andreessen, Venture Capitalist, 2011

"Open source is eating software faster than software is eating the world."

- Joseph Jacks, Venture Capitalist, 2022

Open-Source Software is Non-trivial

- **96.3%** of the world's top one million web servers; **90%** of all cloud infrastructures; **71.65%** of cellphones; **70-90%** of any given software code base.

- Notable examples: Chromium, Python, Linux, Android, Bitcoin, Ethereum, Solana, Meta Llama3, Google AlphaFold

- The Biden administration's Securing Open-Source Software Act of 2022 publicly recognized open-source software as critical economic and security infrastructure.

Example of GitHub Firm Profile

aws

Type / to search

Overview | Repositories 540 | Projects 21 | Packages | People 172 | Sponsoring 2

n April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

Amazon Web Services

Verified | Sponsor

Follow

15.2k followers | United States of America | https://amazon.com/aws | open-source-github@amazon.com

Pinned

aws-cli Public
Universal Command Line Interface for Amazon Web Services
Python 16.9k 4.5k

serverless-application-model Public
The AWS Serverless Application Model (AWS SAM) transform is a AWS CloudFormation macro that transforms SAM templates into CloudFormation templates.
Python 9.6k 2.5k

aws-cdk Public
The AWS Cloud Development Kit is a framework for defining cloud infrastructure in code
TypeScript 12.7k 4.5k

amazon-sagemaker-examples Public
Example Jupyter notebooks that demonstrate how to build, train, and deploy machine learning models using Amazon SageMaker.
Jupyter Notebook 10.9k 7k

s2n-tls Public
An implementation of the TLS/SSL protocols
C 4.7k 776

aws-sdk-go-v2 Public
AWS SDK for the Go programming language.
Go 3.5k 773

People

View all

Sponsoring

Top languages

Python Go Java TypeScript C#

Example of GitHub User Profile

alexzielenski

Type / to search

Overview | Repositories 71 | Projects | Packages | Stars 319

Fork

Popular repositories

Mousecape — Public
Cursor Manager for OSX
Objective-C 1.4k 168

optool
Command Line Tool for interacting with MachO bina
Objective-C 1.2k 231

Star (watch)

ThemeEngine
OS X App to edit compiled .car files
Objective-C 748 130

ZKRevealingTableViewCell
A Sparrow-style Implementation of Swipe-To-Reveal
Objective-C 339 53

StartAtLoginController — Public
A class that uses the new ServiceManagement api to allow apps to run at login.
Objective-C 207 24

ZKSwizzle
Streamlined way to swizzle Objective-C instance and
Objective-C 142 27

Employer

Alex Zielenski
alexzielenski

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@openai

625 followers · 19 following

OpenAI
San Francisco, CA
15:16 - 3h behind
https://alexzielenski.com

Contribution History

184 contributions in the last year

Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct
Mon
Wed
Fri

Projects with the Data

- **Employee Effort and VC Investment** (with Rachel Xiao, Tieying Yu, Metin Sengul), Working paper.
- Conti, Annamaria, Christian Peukert, and Maria P. Roche. "**Beefing IT Up for Your Investor? Open Sourcing and Startup Funding: Evidence from GitHub**." Organization Science 36, no. 4 (July–August 2025): 1551–1573.
- Gupta, Abhinav and Nishesh, Naman and Simintzi, Elena, **Big Data and Bigger Firms: A Labor Market Channel, Working paper.**
- Mei, Kevin, **Why Do Firms Engage in Open-Source Software Development?** Working paper.

The Timeline

- See Professor Itzkowitz's slides.